CardDemo: the demo to download

Quick start for Windows and Linux

Illia Burov

What this is

CardDemo is a sample credit-card system that Amazon Web Services wrote for an IBM mainframe in COBOL: night-time **batch jobs** (posting the day's card transactions, interest, customer statements, a transaction report) and **online screens** for staff (sign-on, accounts, cards, bill payment). The whole system was translated into the modern programming language Rust with Claude Code; the translation shows the same screens and produces the same files as the original.

This download is that translation as **one program** with the sample data. It needs **no installation, no mainframe and no COBOL**: unpack it and start it. It runs on **Windows** (10 or 11, 64-bit) and **Linux** (64-bit, glibc 2.34 or newer: Ubuntu 22.04, Debian 12, Fedora 35, RHEL 9 or later).

Download it from the project's GitHub page, section **Releases**: https://github.com/mehsayer/aws-mainframe-modernization-carddemo-rs-demo/releases

File	For
`carddemo-<version>-windows-x86_64.zip`	Windows
`carddemo-<version>-linux-x86_64.tar.gz`	Linux

Windows: start it with a double click

1. **Unpack the zip file**: right-click it in Explorer, *Extract All…*, *Extract*. (Starting the program from inside the zip view does not work: Windows unpacks only the program, not the sample data next to it.)
2. Open the unpacked folder `carddemo-<version>-windows-x86_64` and **double-click `carddemo.exe`**.
3. The first time, Windows may say *"Windows protected your PC"*, because the program is not signed by a publisher. Click **More info**, then **Run anyway**.
4. A console window opens with the **demo menu**. Type the number of an option and press **Enter**.

The program keeps its data in the folder `carddemo-data` next to `carddemo.exe`. To start again from the original sample data, use option 5 or delete that folder.

Linux: start it in a terminal

```
tar xzf carddemo-<version>-linux-x86_64.tar.gz
cd carddemo-<version>-linux-x86_64
./carddemo
```

The demo menu appears in the terminal. Make the window at least 80 columns by 26 lines (the online screens are 80 by 24 characters, like a mainframe terminal).

The demo menu

Option	What it does	What to look at
1 Online system	The staff screens, as on a 3270 mainframe terminal	Sign on, look up an account, pay a bill (below)
2 Nightly batch	Runs the night's jobs: backup, posting, interest, statements	Each step with its return code; `MAXCC=0000` means success
3 Transaction report	The transaction report for a date range (default: 2022-01-01 to today; run option 2 first)	Then option 4, file `TRANREPT`: transactions per card and account, with totals
4 Output files	A viewer for the files the jobs wrote	Customer statements (text, or HTML in the web browser), rejected transactions
5 Reset	Back to the shipped sample data; removes earlier output files	
6 Original COBOL screens	Linux only, and only if GnuCOBOL is installed (below)	The same screens, run by the original COBOL programs
0 Quit		

A short walk through the online screens (option 1)

Keys: **Enter** sends the screen, **F3** goes back, **F7/F8** page up and down, **F5** saves, **Tab** moves to the next field, **Esc** clears the screen, **Ctrl+Q** returns to the demo menu. A field of full length moves on to the next field by itself, as on a mainframe terminal.

1. User `USER0001`, password `PASSWORD`, **Enter**: the **Main Menu**. (`ADMIN001` / `PASSWORD` signs on as administrator, with the user-management menu.)
2. Type `1` and **Enter**: **Account View**. Account `00000000001`, **Enter**: balance, credit limits, the customer's name and address.
3. **F3** back to the menu. `3` and **Enter**: the **Card List**; **F8** shows the next page. Type `S` next to a card and press **Enter** for its details.
4. **F3** twice, then `10` and **Enter**: **Bill Payment**. Account `00000000001`, **Enter** shows the balance; type `Y` and **Enter** to pay it. The screen shows the new transaction number.
5. **F3** to the menu, **F3** again to sign off, **Ctrl+Q** back to the demo menu.

The nightly batch and its output (options 2 and 4)

Option 2 prints one line per job step: the job (for example `POSTTRAN`), the step, the program (`CBTRN02C`, the original COBOL program's name) and its return code. Then open option 4:

- `STATEMNT.PS`: the customer statements as text; `STATEMNT.HTML` the same as web pages (the viewer offers to open it in the browser).
- `DALYREJS`: the daily transactions that posting rejected, with the reason code and text at the end of each record (for example `0102OVERLIMIT TRANSACTION`).
- `TRANREPT` (after option 3): the transaction report. The nightly batch stamps the transactions it posts with the day it runs, so the report's default range ends today.
- The other files are working files of the jobs, shown as fixed-length records: dots stand for bytes that are not text (packed numbers).

Viewer keys: arrow keys, **PgUp/PgDn**, **Home/End** to scroll, **Left/Right** for wide records, **q** or **Esc** to go back.

Linux: the original COBOL screens side by side (optional)

The Linux download also contains the original COBOL screen programs, compiled with GnuCOBOL 3.2 (folder `cobol-modules`). With the GnuCOBOL 3.2 runtime library installed (`libcob.so.4`, for example from the `gnucobol` or `gnucobol3` package of your distribution, or built from source), menu option **6** runs them on the

same kind of stand-in for the mainframe's transaction system (CICS) that the Rust version uses. Start the program twice, in two terminal windows side by side, and do the same steps with option 1 in one and option 6 in the other: the screens are the same. If the runtime is missing or of another version, the menu says why option 6 is not available. (Their output and messages go to `carddemo-data/cobol-sysout.log`.)

The complete proof, which compiles and runs all original COBOL programs and compares every output file and screen automatically, needs the full project and the GnuCOBOL compiler; see `docs/DEMO.md` in the repository.

If something goes wrong

What you see	What to do
The window closes at once (Windows)	The zip was not unpacked, or `data\EBCDIC` is missing: unpack the whole zip and start `carddemo.exe` from the unpacked folder
"The sample data was not found"	The folder `data/EBCDIC` must be next to the program
Garbled or cut-off screens	Make the window larger (at least 80 x 26)
"Error: … carddemo-data …"	The folder is not writable: unpack the download into your own folder (Downloads, Desktop)
Linux: `GLIBC_2.34 not found`	The system is older than the minimum (glibc 2.34)
The data looks wrong after experiments	Option 5 resets it

The program, its sources and the sample data are under the Apache License 2.0 (files `LICENSE` and `NOTICE`); the sample application CardDemo is © Amazon.com, Inc. or its affiliates.